

“PAPER_{0:D3}” -f [int]# PAPER #78 □ Extragalactic Physics: NED Multi-Wavelength + UQFF

Title: NED Multi-Wavelength Extragalactic Physics: AGN Luminosity Functions and UQFF Buoyancy-Modified Hubble Tension Analysis

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$)

Date: March 7, 2026

Source Data: QCalc_validation.py (DataSourceURLs: NED_BASE, NED_API, QUASAR_SDSS)

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Index Slot: §1.10 Database Integration & Multi-Wavelength Astrophysics, PAPER_078

Abstract

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UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. UQFF Hubble Constant Analysis

Hubble Tension Context

Measurement	H_0 (km/s/Mpc)	Method
Planck 2018 (CMB)	67.4 ± 0.5	Early universe
SH0ES 2023 (Cepheids)	73.0 ± 1.0	Distance ladder
Tension	4.2s	□

UQFF Buoyant Correction to H0

The UQFF vacuum buoyancy modifies the effective expansion rate:

$$H_{\text{UQFF}}(z) = H_0 \times \sqrt{\Omega_\Lambda + \Omega_m(1+z)^3 + [UA] \times \rho_{\text{vac, (UQFF)}} \times 8\pi G/3H_0^2}$$

The [UA] = 0.0001 fractional vacuum coupling adds:

$$\Delta H_0 = H_0 \times [UA] \times 0.5 = 67.4 \times 0.0001 \times 0.5 = 0.0034 \text{ km/s/Mpc}$$

UQFF correction to Hubble tension: $\Delta H_0 = 0.003 \text{ km/s/Mpc}$ – far too small to resolve the 5.6 km/s/Mpc tension. The UQFF does not attempt to resolve Hubble tension through the basic [UA] coupling; a higher-order Resonant Hubble correction would require additional development.

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The UQFF Superconductive mode modifies the AGN accretion efficiency, shifting the break luminosity L_* :

$$L_*^{\text{UQFF}} = L_*^{\text{standard}} \times (1 + [SCm]) = L_* \times 1.99$$

NED quasar catalog comparison:

Redshift bin	L_* _standard ($L_?$)	L_* _UQFF ($L_?$)	NED data range
$z = 0.5$	$10^{\{45.0\}}$	$10^{\{45.3\}}$	$10^{\{44.8\} \pm 10} \{45.5\}$
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UQFF L_* shift of 0.3 dex lies within the 0.5-dex observed scatter – compatible with NED quasar data at all redshifts.

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DLA (Damped Lyman- α) systems contain high column density neutral hydrogen ($N_{\text{HI}} > 2 \times 10^{20} \text{ cm}^{-2}$). The UQFF predicts no modification to the HI 21 cm line frequency (only gravitational Doppler at 10^{-10} level). NED DLA catalog: UQFF-consistent.

Summary

Observable	NED Data	UQFF Prediction	Agreement
H0 tension	5.6 km/s/Mpc	?H0 = 0.003 (negligible)	Not resolved
AGN L*	gap		
	$10^{45} \pm 10^{46.5}$	+0.3 dex ([SCm])	Within scatter
DLA HI column	N_HI > 10^{20}	Unmodified	Compatible

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The NED (NASA/IPAC Extragalactic Database) multi-wavelength catalog covers UV through radio for >1 billion extragalactic objects. Key physics tests for UQFF: (1) AGN luminosity functions comparing UQFF-enhanced accretion vs standard models, (2) the Hubble tension ($H_0 = 67 \pm 7.3 \text{ km/s/Mpc}$) examined through the UQFF Buoyant vacuum correction, and (3) quasar absorption line systems (DLA/LLS) testing the UQFF vacuum density at cosmological redshifts.

UQFF Discovery: Novel application of UQFF calibration constants ($\Delta = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. UQFF Hubble Constant Analysis

Hubble Tension Context

Measurement	H_0 (km/s/Mpc)	Method
Planck 2018 (CMB)	67.4 ± 0.5	Early universe
SH0ES 2023 (Cepheids)	73.0 ± 1.0	Distance ladder
Tension	4.2s	Δ

UQFF Buoyant Correction to H0

The UQFF vacuum buoyancy modifies the effective expansion rate:

$$H_{\text{UQFF}}(z) = H_0 \times \sqrt{\Omega_\Lambda + \Omega_m(1+z)^3 + [UA] \times \rho_{\text{vac, (UQFF)}} \times 8\pi G/3H_0^2}$$

The [UA] = 0.0001 fractional vacuum coupling adds:

$$\Delta H_0 = H_0 \times [UA] \times 0.5 = 67.4 \times 0.0001 \times 0.5 = 0.0034 \text{ km/s/Mpc}$$

UQFF correction to Hubble tension: $\Delta H_0 = 0.003 \text{ km/s/Mpc}$ is far too small to resolve the 5.6 km/s/Mpc tension. The UQFF does not attempt to resolve Hubble tension through the basic [UA] coupling; a higher-order Resonant Hubble correction would require additional development.

2. AGN Luminosity Function Comparison

The UQFF Superconductive mode modifies the AGN accretion efficiency, shifting the break luminosity L_* :

$$L_*^{\text{UQFF}} = L_*^{\text{standard}} \times (1 + [SCm]) = L_* \times 1.99$$

NED quasar catalog comparison:

Redshift bin	L_* _standard ($L_?$)	L_* _UQFF ($L_?$)	NED data range
$z = 0.5$	$10^{\{45.0\}}$	$10^{\{45.3\}}$	$10^{\{44.8\} \pm 10} \{45.5\}$
$z = 1.0$	$10^{\{45.5\}}$	$10^{\{45.8\}}$	$10^{\{45.2\} \pm 10} \{46.0\}$
$z = 2.0$	$10^{\{45.8\}}$	$10^{\{46.1\}}$	$10^{\{45.5\} \pm 10} \{46.3\}$
$z = 3.0$	$10^{\{46.0\}}$	$10^{\{46.3\}}$	$10^{\{45.7\} \pm 10} \{46.5\}$

UQFF L_* shift of 0.3 dex lies within the 0.5-dex observed scatter is compatible with NED quasar data at all redshifts.

3. Quasar Absorption Line Systems

DLA (Damped Lyman- α) systems contain high column density neutral hydrogen ($N_{\text{HI}} > 2 \times 10^{20} \text{ cm}^{-2}$). The UQFF predicts no modification to the HI 21 cm line frequency (only gravitational Doppler at 10^{-10} level). NED DLA catalog: UQFF-consistent.

Summary

Observable	NED Data	UQFF Prediction	Agreement
H0 tension	5.6 km/s/Mpc gap	?H0 = 0.003 (negligible)	Not resolved
AGN L*	$10^{45} \pm 10^{46.5}$	+0.3 dex ([SCm])	Within scatter
DLA HI column	N_HI > 10^{20}	Unmodified	Compatible

Source: `QCalc_validation.py NED_BASE endpoint` | ? = 0.0005/day | [SSq] = 0.57
`.Groups[1].Value "PAPER_{0:D3}" -f $n`

Abstract

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Redshift bin	$L^*_{\text{standard}} (L^?)$	$L^*_{\text{UQFF}} (L^?)$	NED data range
$z = 0.5$	$10^{\{45.0\}}$	$10^{\{45.3\}}$	$10^{\{44.8\} \pm 10^{\{45.5\}}}$
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Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
$\hat{\Gamma}^0$	$5.0 \times 10^{-6} \text{ day}^{-1}$	UQFF exponential decay rate
$[SSq]$	0.57	Universal Quantized Factor
$\hat{\Gamma}^2_i$	0.60 ± 0.61	Buoyancy coupling coefficient
k_{DPM}	1.5	Ug1 DPM-dipole coupling
k_{out}	1.2	Ug2 outer-bubble charge coupling
k_{str}	1.8	Ug3 string-rotation coupling
k_{vac}	2.0	Ug4 vacuum-concentration coupling
$\hat{I} \cdot$	$10 \times 10^{-2} \text{ J}$	Inertia tensor scale
$E_{\text{react}}(0)$	$10 \times 10^{-2} \text{ J}$	Reference reactive energy

A.2 F_U Master Equation (Complete $\hat{\Gamma}^0$ 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 i g l [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}} i g r]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$\hat{\Gamma}^0$	Ug4 dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\hat{I}_{\text{c}}=10\hat{\mu}\gg\hat{A}^1\hat{\mu}^\circ$, $\hat{I}_{\text{a}}=10\hat{\mu}\gg\hat{A}^1\hat{A}^2$, $\hat{I}_{\text{b}}=10\hat{\mu}\gg\hat{A}^1\hat{A}^1$, $\hat{I}_{\text{d}}=10\hat{\mu}\gg\hat{A}^1\hat{A}^3$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imesigl}(1 + 10^{13} \Theta(ho_{SCm} - ho_c)igr) \text{imesigl}(1 + A_q \cos(\Delta\omega t)igr)$$

Symbol	Value	Description
$\hat{I}_{\text{c}}$	$10\hat{A}^1\hat{\mu}\text{ kg/m}\hat{A}^3$	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\hat{I}_{\text{d}}$	$2\hat{I}_{\text{d}}/(434\hat{A}\cdot365.25)\text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	$Ug_{\text{sum}} + \text{Newtonian base}$	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, $\hat{\mu}^\circ$)	Multi-scale field interactions
Buoyant	$\hat{I}_{\text{c}}^2 \hat{I}_{\text{d}} \hat{A} \hat{I}_{\text{b}}$	Expanding nebulae, stellar winds
Superconductive	$\hat{I}_{\text{c}} \hat{A} (1 + 10\hat{A}^1\hat{A}^3\hat{A}\cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, Condensed-Physics.py, and CondensedPhysics2.py.